LGU Exam Department Attendance App

Version 1 Project Report, Lahore Garrison University, Examination Department, 5 November 2026

Status: verified to a testable state. Not live yet. No build, no Play Store publish, no GitHub push, and no paid account was used.

1. Summary

This report covers version 1 of the LGU Exam Department Attendance App, built for the Examination Department of Lahore Garrison University.

Version 1 is verified to a testable state. It is not live yet. No build, no Play Store publish, no GitHub push, and no paid account was used.

The app records staff attendance on each employee's own phone, with a fingerprint check and a 100 metre office radius check, and the HOD downloads the monthly Excel report from a web admin panel.

Item	Status
Employee mobile app (Expo, TypeScript)	Built, typecheck clean
HOD admin panel (Vite React, TypeScript)	Built, production build clean
Database schema (Supabase Postgres, RLS)	Built, reviewed, seed file included
Monthly Excel export	Built, tested on demo data
Leave and absent	Built
Demo data (1 HOD, 3 staff, Nov 2026)	Included in supabase/seed.sql
APK / Play Store	Not built. Needs user approval.
Live test on a real phone	Not done yet. Needs a Supabase project and a phone.

2. Defaults used for v1

The user said "ok start" on 5 November 2026, so v1 was finished with these defaults. Every one of them is editable in the admin panel Settings page. The real values must replace them before go live.

Setting	Default
Office	Examination Department, Lahore Garrison University, Main Campus area, Sector C, DHA Phase 6, Lahore
Office pin (PLACEHOLDER)	31.4640, 74.4427

Setting	Default
Radius	100 metres
Office hours	09:00 to 17:00 PKT
Late after	09:15
Working days	Monday to Friday
Leave allowance	20 days per year, same for all staff

3. What was built: employee mobile app

Expo React Native, TypeScript. One codebase for Android and iOS. Portrait only. LGU green and white. The employee signs in once with their own account and stays signed in on their own phone.

Screen	What it does
Login	Sign in with department email and password. One account per employee.
Home (profile and today)	Shows name, ID, designation, today check in, check out, total hours, and status. Check In and Check Out buttons with fingerprint. Leave balance card. Recent 14 days.
Check In and Check Out	On the Home screen by design. GPS radius check first, then fingerprint, then the server records the time. One check in and one check out per day.
Apply for Leave	Leave type (Casual, Sick, Annual, Emergency, Other), start and end dates, working day count, reason, optional note. Overlaps with a pending or approved leave are refused.
My Leaves	Leave balance for the year and all leave requests with Approved, Pending, or Rejected status and the HOD note.
History	Last 90 days with Present, Late, Leave, and Absent badges and total hours.

4. What was built: HOD admin panel

Vite React, TypeScript. Web app for the HOD only. Role is checked in the app and enforced by database rules. Hosted free on Netlify or Cloudflare Pages. Vercel is not used for this project.

Page	What it does
Today (Dashboard)	Present, Late, On Leave, and Absent lists with counts, checked out count, full table for today, and a Mark absent for today button. Weekend days are labelled non-working.
Leave Requests	Pending requests first. Approve writes one Leave row per working day and reports skipped days. Reject needs a reason, which the employee sees.
Employee Detail	Search any employee, see full history, leave requests, leave balance, pick a month, download their Excel file.
Monthly Report	Pick a month, see the whole department with Present, Late, Absent, Leave, leave taken, leave balance, and total hours per employee, download the full Excel file.

Page	What it does
Employees	Add employees and deactivate them. Deactivation never deletes history. There is no hard delete button in v1.
Settings	Office name, latitude, longitude, radius, start time, end time, late cutoff, working days, and annual leave allowance. A change takes effect on the next check in.

5. Database

Supabase Postgres with Row Level Security. Employees see only their own rows. Only the HOD sees all rows and changes settings. Timestamps come from the database server, never from the phone clock.

Table	Purpose
employees	One row per staff member, linked to one Auth user. Role hod or employee. Deactivation flag, never a delete.
attendance_records	One row per employee per day. Unique key on employee plus date enforces one check in per day. Check in and check out times, GPS distance, location verified flag, mock location flag, device ID, status, total hours (computed by a trigger).
leaves	Leave requests with type, dates, reason, note, status Pending, Approved, or Rejected, and the HOD decision.
office_settings	Single row. Office pin, radius, hours, late cutoff, working days, leave allowance.
device_bindings	First phone an employee signs in on. A new device is recorded so the HOD can see it.
admin_audit	Every HOD action: add employee, change settings, approve or reject leave, deactivate.

6. How a check in works

1. The employee opens the app and taps Check In.

2. The app reads the phone GPS and computes the distance to the office pin with the haversine formula. Over 100 metres, the check in is refused and the distance is shown.

3. The app asks for the phone fingerprint (or face, if the phone uses face unlock).

4. On success, the server records the check in time. A second check in the same day is refused by the database.

5. Check Out works the same way: GPS plus fingerprint, once per day, and the server computes total hours.

6. Check in after 09:15 is marked Late. Check in on or before 09:15 is marked Present.

7. Fingerprint and GPS: honest limits

These limits must be understood before the app is promised to the HOD.

Point	Detail
Fingerprint	Android and iOS do not let any app see or store a fingerprint. The phone only answers yes or no: the person holding it is the device owner. There is no fingerprint database in this app.
Proxy prevention	Comes from four layers together: one login per employee, device binding, GPS radius, and server time. It does not come from fingerprint alone.
What it cannot stop	An employee handing their own unlocked phone to a colleague who is standing inside the office with them. No phone app can stop that. A future version could add a selfie check or a door QR code.
GPS spoofing	The app refuses a reading when the phone reports a mock location, and logs it. This catches casual fake GPS apps. It does not catch a rooted phone with a system level spoofer that hides the flag.
Offline	Check in needs internet at the moment of tapping, because the server timestamp is the record. There is no offline mode in v1.

8. Leave and absent

Leave and absent are part of v1, not an add on.

Rule	Detail
Apply	The employee applies in the app with type, dates, reason, and an optional note. Dates must be real calendar dates in YYYY-MM-DD form.
Approve	The HOD approves in the admin panel. One Leave row is written per working day. A day the employee already checked in on stays Present or Late and is reported as skipped.
Reject	The HOD must write a reason. The employee sees it in the app.
Absent	A working day with no check in and no approved leave is Absent. The HOD taps Mark absent for today after office hours. There is no automatic background job on the Supabase free tier, so v1 does not claim marking runs itself. Until the HOD marks it, the dashboard shows the employee as Absent (not marked yet).
Leave balance	Allowance minus approved Leave days taken in the calendar year, counted in working days only. Default 20 days per year for all staff.
Check in on leave	Refused. An employee on approved leave cannot also record a Present day.

9. Excel reports

Both files are generated in the admin panel from live database data with SheetJS. Times are in Asia/Karachi. Weekends are excluded.

File	Contents
LGU-Attendance--.xlsx	One employee, one month. Columns: Name, Employee ID, Designation, Date, Check In, Check Out, Total Hours, Status, Leave Type, Location Verified.

File	Contents
LGU-Attendance-Full-.xlsx	Every employee, whole month, same columns, plus a Summary sheet with Present, Late, Absent, Leave, Leave Taken (year), Leave Balance (year), and Total Hours per employee.
Sample file	sample/LGU-Attendance-Full-2026-11-DEMO.xlsx in the project, generated from the demo seed data with the same columns. It is a sample, not a file produced by the admin panel.

10. Verification and tests, 5 November 2026

Verification was run on 5 November 2026. Full detail is in TEST_LOG.md.

Check	Result
Mobile TypeScript typecheck (npx tsc --noEmit)	Pass, 0 errors
Admin production build (npm run build)	Pass, dist built, preview serves HTTP 200
Core logic tests (node tests/logic-tests.mjs)	43 passed, 0 failed
Excel builder tests (vitest, real builder code)	5 passed, 0 failed
Em-dash scan (md, ts, tsx, sql, json, css)	0 matches
Sample Excel read back (openpyxl)	63 rows, 2 sheets, status counts Present 10, Late 1, Leave 2, Absent 50

11. Fixes made during verification

Twelve issues were found and fixed in this pass. The first three would have broken a fresh install or a first build.

No.	Fix
1	mobile/App.tsx: the type alias View clashed with the imported React Native View component and failed typecheck. Renamed to ScreenView.
2	admin/src/vite-env.d.ts: added (was missing), so import.meta.env typechecks and the admin build passes.
3	supabase/schema.sql: the ALTER adding leave_id referencing public.leaves ran before the leaves table was created, so a fresh install failed. The leaves table is now created first.
4	supabase/schema.sql, approve_leave: a day marked Absent first and then approved as leave stayed Absent. It is now converted to Leave.
5	mobile/src/lib/leaves.ts: leave dates are now validated as real calendar dates in YYYY-MM-DD form. 2026-02-30 is refused.

No.	Fix
6	admin/src/screens/Settings.tsx: working days are now editable in the form, as the spec already claimed.
7	admin/src/screens/Dashboard.tsx: a weekend day is now labelled as a non-working day.
8	admin/src/screens/Employees.tsx: warns that adding an employee signs the browser in as the new employee, so the HOD signs back in after.
9	admin/src/App.tsx: the HOD role is now rechecked on every auth state change, not only on first load.
10	mobile/assets/icon.png and splash.png: created (simple LGU green placeholder branding). app.json referenced files that did not exist.
11	Office placeholder pin corrected to 31.4640, 74.4427 (public coordinates for LGU Main Campus). The old pin was about 3 km off. It is still a placeholder.
12	admin/src/lib/excel.test.ts: the test helper typed its leave type as LeaveType with a null default, which failed strict typecheck. The final build sweep caught it. Fixed to LeaveType | null, build and tests rerun clean.

12. What is NOT done, and what is NOT tested yet

No EAS build was run. No APK exists yet. No GitHub push was made. No Play Store listing exists. No paid account was created.

Item	Detail
Live database	No live Supabase project exists for this app yet, so login, check in with a real phone, fingerprint, GPS, leave approval, and the mark absent function have not run against a real database. The SQL was reviewed and the logic was tested, but a live run is still needed.
Phone test	Fingerprint and GPS do not work properly in a simulator. The first real test must be on a real Android phone at the office.
APK	Needs user approval first. When approved: push to GitHub first, then run the EAS build.
Play Store	Needs a one time $25 Google developer account fee. The free v1 route is a direct APK download link.
iOS	No App Store build in v1. It needs an Apple Developer account at $99 per year.
Notifications	No push notifications in v1. No forgotten check out reminder and no leave request alert yet.
Payroll	Total hours are reported. Pay is not computed.

13. What is still needed from the user

Nothing in this list blocks the code. All of it blocks go live.

Item	Detail
Exact office pin	Latitude and longitude of the Examination Department office door. Stand at the door, open Google Maps, long press the pin, copy the numbers. Or send the Maps pin. The placeholder is the campus area only.
Staff roster	Full list: full name, employee ID, designation, official email, phone. A template is in staff-roster-template.csv. Demo staff (EX-001 to EX-003) must be replaced.
Office timings	Confirm start 09:00, end 17:00, Late after 09:15, Monday to Friday. Does the department work Saturdays? Is there a holiday list?
HOD details	Full name and email for the admin login.
Leave allowance	Confirm 20 days per year for all staff, or give separate numbers for casual and sick leave, or different numbers per designation.
LGU logo	Official logo file (PNG with transparent background) to replace the placeholder icon and splash. The LGU Exam Manager desktop app has lgu-logo.jpg; confirm whether it may be reused.

14. Next steps

1. User approves the next step.

2. Create the free Supabase project and run supabase/schema.sql, then supabase/seed.sql for testing.

3. Create the HOD Auth account and link the demo staff, or create staff from the admin panel.

4. Test on a real Android phone at the office: sign in, check in with fingerprint inside the radius, check in refused outside the radius, check out, apply for leave, HOD approves, download the monthly Excel.

5. Push to GitHub first, then run the EAS APK build, only after the user approves the build.

6. Share the free APK download link with staff. Play Store later, if the $25 fee is paid.

7. Replace the demo roster and the placeholder pin with the real roster and the real office pin, remove the demo rows, and set the real HOD account.